FERDIANSYACH

Full-Stack Developer (React/Next.js, Node.js) | ML for Energy Forecasting

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PROFESSIONAL SUMMARY

Full-Stack Developer with hands-on experience architecting and shipping production-grade web applications using React, Next.js, and Node.js across the full frontend-to-backend stack. Delivered 5+ production projects achieving Lighthouse scores of 90+ and engineered a predictive AI model with 92% accuracy processing 50,000+ data points at Telkom Indonesia. Leverages Python-based ML expertise (LSTM, XGBoost) as a data-driven catalyst to build smarter, more impactful software products.

PROFESSIONAL EXPERIENCE

Business Intelligence & ML Intern
Apr 2025 – Sep 2025

Telkom Indonesia — Infranexia | Internship | Jakarta, ID

- Engineered a predictive AI model using LSTM and Gradient Boosting that achieved 92% accuracy in smart building energy consumption forecasting, directly supporting the techno-economic analysis division.
- Architected an end-to-end data pipeline from raw ingestion through preprocessing to model deployment (LSTM, XGBoost), processing 50,000+ data points and generating actionable insights for energy efficiency optimization.
- Developed a web-based internship management platform in collaboration with 3 cross-functional divisions, serving 100+ interns and reducing onboarding coordination overhead.

Computer Laboratory Assistant
Sep 2024 – Sep 2025

Lab Data Monetize — Universitas Nasional | Student Organization | Jakarta, ID

- Maintained and troubleshot 30+ laboratory computer units, sustaining 98% device availability during operational hours and minimizing session downtime for 200+ students per semester.
- Optimized software installation and system configuration workflows, supporting lab sessions for 200+ students per semester through standardized maintenance procedures.

Research & Development Assistant
Jun 2024 – Mar 2025

HIMASI — Universitas Nasional | Student Organization | Jakarta, ID

- Orchestrated a national technology seminar as Chief Executive, coordinating 15 committee members from planning to execution and successfully delivering the event to 150+ attendees on schedule.
- Developed and managed the association's official WordPress website, increasing website traffic by 40% and driving member engagement through structured, SEO-optimized content.

Fullstack Developer
Apr 2024 – Nov 2024

UNAS FEST — Universitas Nasional Festival | Student Organization | Jakarta, ID

- Developed 10+ responsive website components using TypeScript and Tailwind CSS, achieving a Lighthouse score of 90+ for performance and accessibility on the official festival portal.
- Engineered a comprehensive testing and debugging pipeline that reduced the production bug rate by 60%, ensuring optimal stability for the public release used by the entire university community.

Front-End Developer
Aug 2024 – Oct 2024

Anambas Tourism Website | Freelance | Jakarta, ID

- Developed 8+ interactive and responsive UI components for a tourism platform using Next.js, optimizing high-resolution image loading by 50% via Next.js Image optimization and lazy loading.
- Architected a modular front-end component system that significantly improved Core Web Vitals scores, boosting overall page performance and user experience for the tourism platform.

EDUCATION

Universitas Nasional
2022 – 2026

Bachelor of Computer Science (S.Kom) — Information Systems

GPA: 3.77

Capstone Project: Cloud-Based Coastal Water Quality Monitoring: Spatiotemporal Trend Analysis and Unsupervised Spectral Clustering via Web-GIS

Relevant Coursework: Web Programming, Database Systems, Software Engineering, Artificial Intelligence, System Analysis & Design, Computer Networks

TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, HTML/CSS, Tailwind CSS

Backend & DB: Node.js, Express.js, REST API, MongoDB, SQL

Data & ML: Python, LSTM, XGBoost, Pandas, Streamlit

DevOps & Tools: Git, Docker, Google Cloud, Figma, WordPress

SELECTED PROJECTS

Indosaji E-commerce
GitHub ↗

An end-to-end full-stack e-commerce platform based on MERN with payment gateway integration.

`React · Node.js · Express.js · MongoDB · REST API · JWT Auth`

Smart Meter Analysis
GitHub ↗

Predictive AI models (LSTM & XGBoost) for energy anomaly detection with a real-time interactive dashboard.

`Python · Machine Learning · Data Analysis · Streamlit · XGBoost · LSTM`

Anambas Tourism Website

An interactive and responsive tourism platform built with the Next.js and Tailwind CSS ecosystem.

`Next.js · React · Tailwind CSS · TypeScript`

CERTIFICATIONS & LICENSES

Junior Web Developer — BNSP (Badan Nasional Sertifikasi Profesi) (Feb 2026)

IT Specialist — Python — Certiport (Pearson VUE) (Oct 2025)

Internet of Things Fundamental — Telkom Indonesia (Apr 2025)

Google Cloud Computing Foundations — Google Cloud (Dec 2025)

Scrum Mastery — Telkom Indonesia (Jul 2025)

Statistics Using R for Data Science — DQLab (Feb 2023)